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Large language models as internal consistency checkers for ophthalmic manuscripts

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Large language models (LLMs) are increasingly used in scientific writing, while their potential application to editorial assessment and peer review has attracted growing interest [1–3]. However, their ability to identify internal inconsistencies within scientific manuscripts remains poorly characterised. We evaluated four contemporary LLMs as automated consistency checkers using published ophthalmic literature.

Of the 40 included articles randomly selected from ophthalmic journals with an impact factor of >3.0, a prespecified inconsistency was introduced in 30 studies without altering the manuscript. These included: percentage/count ($n = 10$), sample-size/subgrouping ($n = 10$), and statistical reporting inconsistency ($n = 10$). Ten manuscripts remained unmodified. Claude Sonnet 5 High Effort, GPT-5.6 High Effort, GPT-5.5 Instant, and Gemini 3.1 Pro independently reviewed every article using an identical prompt. For planted errors, detection, exact localisation, correct explanation, and replacement value were recorded. All additional errors flagged by LLMs were manually adjudicated as either genuine or non-genuine. Differences in planted-error detection across models were assessed using Cochran's Q test, followed by pairwise McNemar tests with Holm correction for multiple comparisons.

Across 120 assessments of manipulated manuscripts, a total of 98 planted errors were detected (81.7%). Sensitivity was highest for GPT-5.6 (29/30, 96.7%) and Claude Sonnet 5 (27/30, 90.0%), followed by Gemini 3.1 Pro (23/30, 76.7%), and GPT-5.5 (19/30, 63.3%; Cochran $Q = 16.86$, $p < 0.001$) (Table 1). Following Holm correction, GPT-5.6 and Claude Sonnet 5, each outperformed GPT-5.5 ($p = 0.012$ and $p = 0.039$, respectively). Detection was highest for sample-size/subgrouping inconsistencies (36/40, 90.0%), followed by percentage/count (33/40, 82.5%) and statistical reporting inconsistencies (29/40, 72.5%). Once detected, all 98 planted errors were correctly localised and 97/98 (99.0%) were correctly explained and corrected. Complete end-to-end performance (detection, localisation, explanation and correction) was achieved in 97/120 assessments (80.8%).

LLMs also identified substantial numbers of genuine pre-existing inconsistencies. Across all 40 manuscripts, 337/355 additional error flags (94.9%) were verified as genuine, representing 158 unique pre-existing inconsistencies after duplicate detections of the same inconsistency across models were collapsed: 134/139 (96.4%) for GPT-5.6, 89/92 (96.7%) for Claude Sonnet 5, 44/46 (95.7%) for GPT-5.5 and 70/78 (89.7%) for Gemini (Table 2). In manipulated manuscripts, 222/237 additional flags (93.7%) were genuine. In the ten unmodified manuscripts, 115/118 flags (97.5%) represented genuine inconsistencies, with only three non-genuine flags (2.5%). GPT-5.6 identified more additional genuine inconsistencies than Claude Sonnet 5 (134 vs. 89; Holm-adjusted $P < 0.001$), most commonly comprising rounding discrepancies (99/134, 73.9%).

Performance differed principally in error discovery rather than error interpretation. Once an inconsistency was recognised,

Table 1. Detection of prespecified planted inconsistencies.

Model	Overall *n*/30 (%)	Percentage/count *n*/10	Sample-size/ subgrouping *n*/10	Statistical reporting *n*/10	Correct explanation and correction*
GPT-5.6 High Effort	29/30 (96.7)	10/10	10/10	9/10	29/29 (100)
Claude Sonnet 5 High Effort	27/30 (90.0)	10/10	9/10	8/10	26/27 (96.3)
Gemini 3.1 Pro	23/30 (76.7)	7/10	10/10	6/10	23/23 (100)
GPT-5.5	19/30 (63.3)	6/10	7/10	6/10	19/19 (100)
Overall	98/120 (81.7)	33/40 (82.5)	36/40 (90.0)	29/40 (72.5)	97/98 (99.0)

*Among planted errors detected. All 98 detected errors were exactly localised.

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Table 2. Additional error identification across all manuscripts.

Model	Manipulated manuscripts, genuine/flags (%)	Unmodified manuscripts, genuine/flags (%)	False-positive flags in unmodified manuscripts, *n* (%)	All manuscripts, genuine/flags (%)
GPT-5.6 High Effort	90/93 (96.8)	44/46 (95.7)	2/46 (4.3)	134/139 (96.4)
Claude Sonnet 5 High Effort	58/61 (95.1)	31/31 (100)	0/31 (0)	89/92 (96.7)
GPT-5.5	27/28 (96.4)	17/18 (94.4)	1/18 (5.6)	44/46 (95.7)
Gemini 3.1 Pro	47/55 (85.5)	23/23 (100)	0/23 (0)	70/78 (89.7)
Overall	222/237 (93.7)	115/118 (97.5)	3/118 (2.5)	337/355 (94.9)

localisation and correction were almost universally accurate. In contrast, some planted errors were missed despite models identifying other genuine inconsistencies within the same manuscript, indicating that failure of systematic detection remains the principal limitation across current LLMs.

As the applications of LLMs continue to grow, these findings support their use as an additional layer of manuscript quality control prior submission [4, 5]. Their role should be best viewed as complementary, rather than a replacement, with potential value for systematic editorial screening. Automated screening may identify inconsistencies missed during conventional review, potentially reducing routine checking burden and improving reporting quality.

DATA AVAILABILITY

The datasets generated and analysed during the current study are available from the corresponding author on reasonable request.

REFERENCES

1. Lee J, Yoo JJ. The role of large language models in the peer-review process: opportunities and challenges for medical journal reviewers and editors. J Educ Eval Health Prof. 2025;22:4. https://doi.org/10.3352/jeehp.2025.22.4.
2. Paris J, Kleinig O, Agarwal A, Chan WO, Selva D. Performance of large language models for ophthalmic literature retrieval. Eye (Lond). 2026;40:1759–60. https://doi.org/10.1038/s41433-026-04586-y.
3. Li M, Sun J, Tan X. Evaluating the effectiveness of large language models in abstract screening: a comparative analysis. Syst Rev. 2024;13:219. https://doi.org/10.1186/s13643-024-02609-x.
4. Bedi S, Liu Y, Orr-Ewing L, Dash D, Koyejo S, Callahan A, et al. Testing and evaluation of health care applications of large language models: a systematic review. JAMA. 2025;333:319–28. https://doi.org/10.1001/jama.2024.21700.
5. Paris J, Agarwal A, Chan WO, Selva D. Can large language models accurately triage oculoplastic referrals? Eye (Lond). Published online August 7, 2026. https://doi.org/10.1038/s41433-026-04766-w.

AUTHOR CONTRIBUTIONS

Ethan La: methodology, data curation, formal analysis, investigation, writing – original draft. **Jai Paris:** conceptualisation, methodology, data curation, formal analysis, investigation, writing – original draft. **Ayushi Agarwal:** data curation, investigation, interpretation of results, and writing – review and editing. **Weng Onn Chan:** conceptualisation, methodology, supervision, interpretation of results, and writing – review and editing. **Dinesh Selva:** conceptualisation, supervision, interpretation of results, and writing – review and editing.

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